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Advancing Stroke Prediction: A Comparative Study of Hybrid Ensemble and Deep Learning Models

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***Abstract*—Stroke remains a major contributor to global mortality and long-term disability which places a strain on global healthcare systems. This study introduces two advanced hybrid ensemble learning models—the Stacking Classifier and Voting Classifier—for stroke prediction. To ensure model robustness, preprocessing techniques such as SMOTE for addressing class imbalance and feature standardization for numerical stability were implemented. The Stacking Classifier, which combines Random Forest, XGBoost, and LightGBM as base learners with a Logistic Regression meta-learner, achieved exceptional performance, including an accuracy of 96.97%, precision of 97.40%, recall of 96.49%, and a ROC-AUC of 99.57%. Similarly, the Voting Classifier, employing Random Forest, XGBoost, CatBoost, and LightGBM with soft voting, delivered competitive results with an accuracy of 97.02% and a ROC-AUC of 99.60%. Both models consistently outperformed traditional machine learning techniques, standalone deep learning models, and hybrid deep learning approaches. These results establish the proposed models as effective, reliable, and clinically relevant tools for precise and early stroke risk assessment, enhancing patient outcomes and facilitating timely medical interventions.**

Index Terms*—Ensemble machine learning, Stroke prediction,hybrid models, Stacking model, Voting model, Deep belief network,deep learning, hybrid deep learning.

I. Introduction

Stroke is considered one of the leading causes of death and long-term disabilities globally with very high burdens to health care systems according to the WHO. The WHO global report shows that annually 15 million people experience a stroke and death happens every four to five minutes [1]. Both clots (ischemic stroke) and bleeds (hemorrhagic stroke) occur in the brain's blood vessels, cutting off the oxygen supply and causing massive destruction of brain tissue. The hemorrhagic stroke makes up approximately 15% [1]. Long term disabilities tend to result from surviving, such as speech difficulties, memory loss and impaired motor functions that erode much of the quality of their lives. It is also economically costly with many billions being spent every year on a treatment, rehabilitation and long term care.

This is because medical advancements in this particular problem have not kept pace with the broader progress in modern medical science. Stoke risk assessment is important to minimize the frequency of the occurrences and enhance the quality of the clients. Computerized procedures can make use of numerical methods to estimate stroke risk given clinical and demographic data.

II. Related Works

The use of machine learning for brain stroke prediction has seen significant advancements, with various studies proposing innovative models and methodologies to enhance prediction accuracy and reliability.

Uppal et al. [1] aimed to enhance the performance of Multi-Layer Perceptron (MLP) models by experimenting with three optimization techniques: Adadelta, RMSProp, and AdaMax. Testing on different data splits (80:20 and 90:10), RMSProp delivered the best results with 95.8% training and 94.9% testing accuracy, while AdaMax minimized training loss but had slightly lower accuracy.

In [2], a hybrid deep learning model combining Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks was introduced for ischemic stroke prediction. CNN handled feature extraction and LSTM captured temporal dependencies, achieving 95.9% accuracy and 98.9% AUC-ROC. This outperformed traditional models like Logistic Regression, Random Forest, XGBoost, and standalone CNN and LSTM architectures.

Issa et al. [3] proposed an Artificial Neural Network (ANN) for binary stroke classification. Their multi-layered model reached 96.12% accuracy with high sensitivity (97.8%) and specificity (93.72%). However, no comparisons with other machine learning techniques were provided.

A predictive framework was developed in [4] using electronic health records (EHR) to identify key stroke predictors such as age, heart disease, glucose level, and hypertension. Neural networks with selected top features achieved 78% accuracy. The use of PCA-transformed features reduced accuracy slightly, highlighting the trade-offs in dimensionality reduction.

Alruily et al. [5] proposed an ensemble model (RXLM) combining Random Forest, XGBoost, and LightGBM on a Kaggle dataset. Their method included KNN imputation, outlier removal, SMOTE for class balancing, and hyperparameter tuning. The ensemble achieved 95.29% accuracy and

99.13% AUC, outperforming individual classifiers. However, comparisons with advanced ensemble techniques like stacking or hybrid deep learning models were lacking.

Rahman et al. [6] compared various algorithms—RF, XGBoost, AdaBoost, LightGBM, and ANN. The four-layer ANN outperformed other deep learning models, achieving 92.39% accuracy.

Srinivasu et al. [7] developed an interpretable ANN-based heart stroke prediction model, achieving 95% accuracy. The study applied SMOTE+Tomek for class balancing and used explainability tools such as permutation importance and LIME, enhancing model transparency for clinical use.

Balaji et al. [8] proposed a coronary heart stroke prediction system using ML techniques (Logistic Regression, SVM, Naïve Bayes, Random Forest, Decision Tree), achieving 80.29% training and 72.68% test accuracy with a Gradient Boosting Classifier. The authors highlighted the need for advanced feature selection and real-time clinical validation.

Rahman et al. [9] applied ensemble methods—stacking and voting classifiers—for coronary disease prediction. Their stacking model, combining RF, XGBoost, LightGBM with a logistic regression meta-learner, achieved 92.56% accuracy and 97.65% AUC, outperforming traditional models and deep learning approaches.

III. METHODOLOGY

This section illustrates an overview of the steps involved in building and assessing hybrid models for heart disease prediction. At first we describe the dataset and the preprocessing methods which we used to prepare the data. We then explain the architecture of the hybrid models, which combines several classifiers through stacking and voting mechanisms. Finally, the models are evaluated using multiple metrics to ensure their effectiveness and accuracy in predicting outcomes.

A. Dataset

For this study, we used the Fedesoriano Stroke Prediction Dataset from Kaggle [10] which includes detailed clinical and demographic data for 5,110 patients. Each patient record consists of 11 features such as gender, age, hypertension status, heart disease status, marital status, work type, residence type, average glucose level, body mass index (BMI), and smoking status. The dataset's target variable indicates whether a patient has experienced a stroke (1) or not (0).

B. Data Preprocessing

The dataset was preprocessed for both hybrid models in analysis to manage missing data and normalize numerical features, and encode categorical variables. The following steps were performed:

- **Handling Missing Values:** For numerical features, such as BMI, missing entries were imputed using the median strategy to minimize outlier influence. For categorical features, such as smoking_status, missing values were replaced with the most frequent category (mode) to maintain consistency in the dataset.
- **Feature Transformation:** Numerical features like age, avg_glucose_level, and BMI were standardized using StandardScaler. This ensured all features operated on equal scales, improving the model's stability and performance. Categorical features, including gender, work_type, and residence_type, were transformed into binary representations using one-hot encoding, making them suitable for algorithmic processing.
- **Class Imbalance Handling:** The data was unbalanced in that there were less stroke cases as compared to the non-stroke cases. To remedy this, synthetic stroke data samples were created using SMOTE (Synthetic Minority Over-sampling Technique) balancing the dataset and providing a better model with more accurate predictions and capabilities of being used in the detection of stroke events.

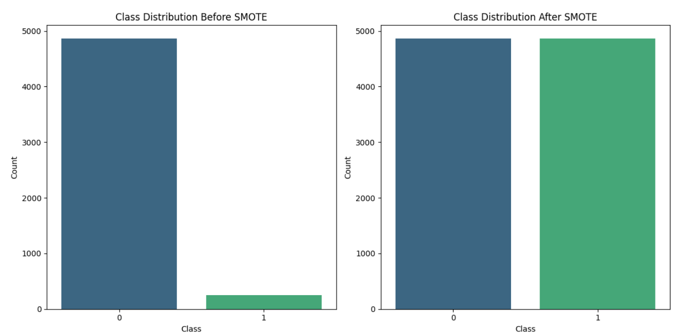

Fig. 1: Class distribution before and after SMOTE balancing.

C. Overview of Hybrid Models

The Stacking Classifier and the Voting Classifier are the two machine learning hybrid models that we suggest in our study regarding stroke prediction. The two methods aggregate several underlying learners in order to increase the accuracy of prediction, robustness and generalization.

D. Stacking Classifier

1) Model Architecture: The stacking classifier is a two-level ensemble learning algorithm which combines various lower level classifiers using an upper level one called a meta-learner. Stacking unlike any other traditional ensemble technique like bagging or boosting, permits miscellaneous models to be combined in a supervised parcel, and provides the meta-learner to be educated to maximize the combo of the results of the fundamental models. This multilayered structure is especially successful at learning a wide range of learning patterns across the models to increase the predictive quality and general stability of the model.

2) Base Learners: Three base learners were chosen in the current research: Random Forest, XGBoost, LightGBM. Every base model was selected due to its proven results in classifying structured data:

- RandomForest Classifier : Random Forest lowers the variance due to averaging of prediction with a number of decision trees trained using bootstrapped samples.

- XGBoost Classifier : The XGBoost is gradient boosting that is regularized and manages complex patterns and outliers effectively.
- LightGBM Classifier : LightGBM is based on tree learning with high-speed using histograms when training on large data.

Each of the base learners was trained on the same dataset separately and provided class probabilities of the output instances. These probabilistic predictions are inputs to the meta-learner and they record the various decision boundaries and confidence levels of each model.

3) Meta-Learner: The architecture employed the meta-learner which is the Logistic Regression model. It trains over the concatenated output of the base learners, to make the final prediction. LR was chosen because of its simplicity and interpretability and good results with respect to classification binary problems. The model is trained to give suitable weights to the prediction made in the base learners, depending on how each one was reliable across the feature space.

For an input instance x, the model aggregates the predictions $f_i(x)$ from each base classifier:

$$\hat{y} = g(f_1(x), f_2(x), f_3(x))$$

Here, g represents the logistic regression function that synthesizes these predictions into a final class label $\hat{y}$.

By employing this stacking method, the model capitalizes on the complementary strengths of the various classifiers, leading to a more robust prediction framework.

4) Training Procedure and Overfitting Prevention: Internal k-fold cross-validation is used to decrease the bias and avoid the overfitting occurs within the StackingClassifier through cv=5. Here the base learners are trained on k-1 folds, and they are employed to make out-of-fold prediction in the rest of k-th folds. All these predictions are pooled together across all the folds and fed to train the meta-learner so as to allow the meta-learner to learn exclusively on the predictions applied on the data not used in training the base models. Such division of training and meta-training helps to improve overgeneralization and minimizes overfitting. The recommended architecture of stacking classifier is depicted in Figure 2.

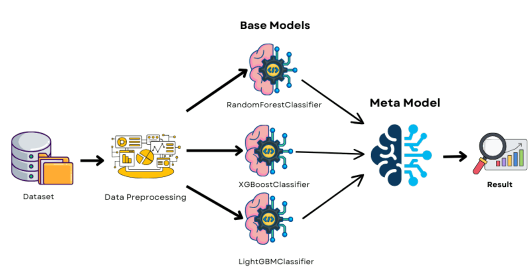

Fig. 2: Workflow of the proposed stacking classifier model

Stacking architecture combines the good features of the models used. Random Forest is tolerant of the noise in the patterns of data and XGBoost and LightGBM show expertise in complex interaction modeling. The meta-learner uses dynamical allocation of explanatory weight to these models thus improving the accuracy of prediction and generalization, which is vital in clinical situations like predicting stroke.

E. Voting Classifier

1) Model Architecture: The Voting Classifier refers to an ensemble architecture where the outputs of a set of diverse base classifiers are combined by aggregating the individual prediction of each model so as to improve classification accuracies as well as the robustness of the models. However, unlike a mechanism based on stacking that makes use of a metaclassifier to determine that one weight of the previous decision rules will be applied, the current method uses a direct prediction combination approach. The ensemble is not fitted in normal way but through soft voting whereby the probabilistic results of the separate models are averaged using operation of a weighted average. The diversity is particularly beneficial in a form of such an arrangement when the learners at the base have complemented decisions and disparate confidence levels.

2) Base Learners: The four majorly used base classifiers (Random Forest, XGBoost, CatBoost, and LightGBM) were chosen to form the ensemble.

- Random Forest : Random Forest is a bagging form of model that can handle noise and variance.
- XGBoost : XGBoost uses gradient boosting, but with the strong regularization and shows good results on the structured data
- CatBoost : CatBoost is an ordered gradient boosting algorithm that can maximise the use of categorical features by reducing overfitting using ordered boosting.
- LightGBM : LightGBM is a primeal gradient-boosting algorithm that uses learning over a histogram in order to achieve maximum speed and scalability.

The same training data was used; each base learner was trained separately. Collectively, all the models returned a probability during prediction and the multiple probabilities were combined through weighted averaging where the weight of the models was set to 1, 2, 1.5, and 1 as the weights of each model Random Forest, XGBoost, CatBoost and LightGBM, respectively. The last ensemble output was the class whose weighted average was maximum probability.

3) Voting Strategy: The Voting Classifier uses soft voting, which simply refers to averaging the predicted probabilities from all classifiers and then picking the class with the highest one. This method allows the model to weigh the certainty of its individual base classifiers to ensure a more reliable and knowledgeable final decision.

In soft voting, each classifier outputs a probability for each class based on its own internal model. The Voting Classifier aggregates these probabilities across all classifiers to form a final prediction. This method is particularly useful when the classifiers exhibit different strengths and weaknesses, as it allows the model to benefit from their combined insights.

In the soft voting method, each base classifier $P_i(y|x)$ outputs a probability distribution over the classes. The final prediction $\hat{y}$ is calculated as:

$$\hat{y} = \arg\max \sum_{i=1}^{n} w_i P_i(y|x)$$

where w_i represents the weight assigned to each base classifier, and $P_i(y|x)$ is the predicted probability from the i-th classifier.

4) Feature Selection and Preprocessing: Preprocessing of the dataset is done on a pipeline designed in the following manner, before any analytical process. The missing numerical variables are imputed with median, whereas the categorical features are filled with mode. Thereafter standardization is carried out so that variables can have similar scale. This is followed by one-hot encoding of categorical features into binary features. Higher-order interactions between variables are then modelled by sequential polynomials expansions. Lastly, an automated procedure, Recursive Feature Elimination with Cross-Validation (RFECV) works on a Random Forest estimator to find out the most informative features. The follwoing architecture of stacking classifier is depicted in Figure 3.

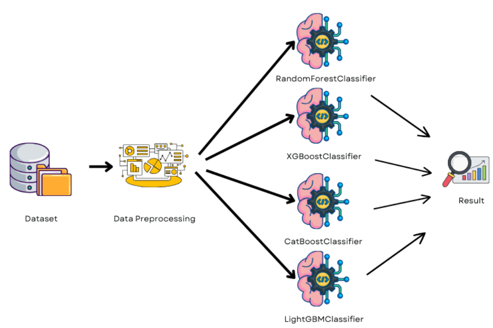

Fig. 3: Workflow of the proposed voting classifier model

Voting Classifier is an efficient method of implementing lots of other strong learners, without a deep-level learner. It brings together ensemble capabilities of all of them, so, it can sum related strengths both of probabilistic abilities of prediction (Random Forest), regularization (XGBoost), support of categorical variables (CatBoost), and speed of computation (LightGBM). The additional robustness of the soft voting is also an advantage particularly when it is supposed to be applied in the conditions of stroke prediction where the false positive rate is required to be kept at minimum and the recall at the maximum. This framework hence combines good performance with interpretability along with computational tractability.

F. Evaluation Metrics

To evaluate the models, we used the following metrics:

- **Accuracy**: This is the ratio of the number of instances we predicted correctly, considering both true positives and true negatives, divided by the total number of instances.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Where:
  - TP = True Positives (correctly predicted positive cases)
  - TN = True Negatives (correctly predicted negative cases)
  - FP = False Positives (incorrectly predicted positive cases)
  - FN = False Negatives (incorrectly predicted negative cases)

- **Precision**: The ratio between the instances correctly labeled as positive (true positives), and instances labeled as positive. This measurement indicates the capability of the model to not make false positive or identify positive cases reliably.

$$\text{Precision} = \frac{TP}{TP + FP}$$

- **Recall**: The probability of the model making right predictions of present positive instances divided by the sum of positive cases present. This is an index which shows the sensitivity at which the model picks the positive results.

$$\text{Recall} = \frac{TP}{TP + FN}$$

- **F1 Score**: Harmonic mean of precision and recall gives an equal value of these measures, that is, the model needs to be good in terms of precision and sensitivity.

$$\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

- **ROC AUC Score**: The Receiver Operating Characteristic curve area measures the capability of the model to classify classes in a good way. The AUC shows how well the positive and the negative classes separate.

$$\text{AUC} = \int_0^1 \text{TPR}(\text{FPR})\, d(\text{FPR})$$

Where:
  - TPR = True Positive Rate (Recall) = $\frac{TP}{TP+FN}$
  - FPR = False Positive Rate = $\frac{FP}{FP+TN}$

Additionally, ROC Curves and Precision-Recall Curves were plotted to further analyze the models' trade-offs between precision, recall, and false positive rates.

IV. Results and Analysis

This study aims to evaluate the performance of two hybrid machine learning models, the Stacking Classifier and the Voting Classifier, for predicting stroke occurrences using the Fedesoriano Stroke Prediction Dataset. The models were assessed using key performance metrics, including Accuracy, Precision, Recall, F1 Score, and ROC-AUC. Both models demonstrated strong predictive performance, with the Stacking Classifier slightly outperforming the Voting Classifier across most metrics.

A. Stacking Classifier Performance

The Stacking Classifier, which combines Random Forest, XGBoost, LightGBM, and CatBoost as base learners with Logistic Regression as the meta-learner, delivered exceptional results. It achieved an accuracy of 96.97%, precision of 97.40%, recall of 96.49%, F1 score of 96.94%, and a remarkable ROC-AUC of 99.57%. These metrics demonstrate the model's ability to accurately detect stroke cases with high sensitivity and a low false positive rate, making it highly suitable for healthcare applications. The high metrics indicates a near-perfect balance between sensitivity and specificity, ensuring reliable predictions. The ROC curve obtained by Stacking Classifier is as shown in the figure 4.

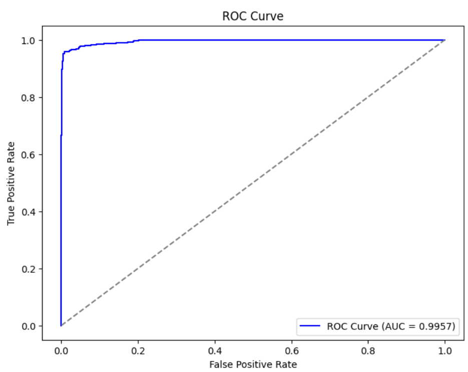

Fig. 4: Workflow of the proposed voting classifier model

B. Voting Classifier Performance

The Voting Classifier, which uses soft voting to combine predictions from Random Forest, XGBoost, LightGBM, and CatBoost, also performed impressively. It achieved an accuracy of 97.02%, precision of 97.20%, recall of 96.80%, F1 score of 97.00%, and a ROC-AUC of 99.60%. The model's high precision highlights its strength in minimizing false positives, making it particularly valuable in scenarios where overprediction carries significant risks. However, its slightly lower recall compared to the Stacking Classifier means it may miss some stroke cases. The ROC curve obtained by Stacking Classifier is as shown in the figure 5.

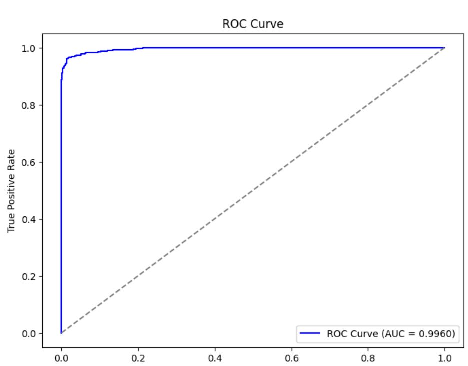

Fig. 5: Workflow of the proposed voting classifier model

C. Performance Comparison: Stacking vs. Voting Classifiers

The comparison of the Stacking and Voting models shows that both of those models demonstrated highly competitive results on all performance metrics, which supports the usefulness of the ensemble learning in the stroke prediction. Stacking Classifier exhibited better precision (97.40%) or lower false positive which is useful in conditions where over diagnosis has to be avoided at all costs. In contrast, Voting Classifier performed slightly better in terms of the accuracy (97.02%), recall (96.80%), F1 score (97.00%), andROC-AUC (99.60%). Such scores indicate that the Voting Classifier demonstrated a better ability to balance sensitivities and specificities.

Soft voting is preferable as it is easy to learn and directly and probabilistically combines the predictions of different base learners, they may take advantage of the strengths of each of the base learners, and do not need a separate meta-model. In the meantime, the Stacking Classifier uses meta-learner to interpret and optimise the outputs of base models, which leads to stable performance and only slightly more complex. The comparison between the two models is showed in the figure 6.

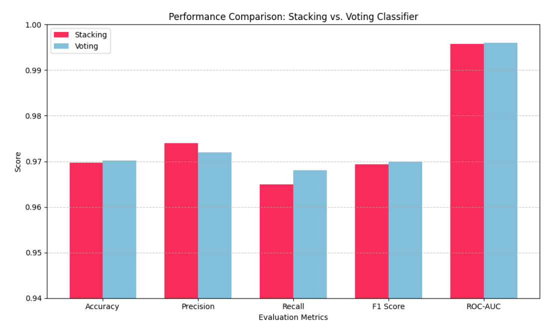

Fig. 6: Workflow of the proposed voting classifier model

D. Comparison with Baseline Models:

Experimental results demonstrated that the proposed hybrid machine learning models, specifically the Stacking Classifier and Voting Classifier, consistently outperformed traditional machine learning models, standalone deep learning models, and hybrid deep learning approaches. This highlights the superior generalization and robustness of ensemble learning methods for stroke prediction, with a detailed comparison of these models presented in Table 1.

TABLE I: Model Performance Metrics

Model	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)	ROC-AUC (%)
Voting Classifier	**97.02**	**97.20**	**96.80**	**97.00**	**99.60**
Stacking Classifier	**96.97**	**97.40**	**96.49**	**96.94**	**99.57**
Decision Tree (DT) [4]	74.00	75.00	–	–	–
Random Forest (RF) [4]	74.00	74.00	–	–	–
Convolutional Neural Net (CNN) [4]	74.00	74.00	–	–	–
Artificial Neural Network (ANN) [3]	91.78	95.69	97.80	96.73	–
Hybrid CNN+LSTM [2]	95.90	–	–	–	98.90
MLP with RMSProp Optimization [1]	94.98	–	–	–	–
MLP with AdaMax Optimization [1]	94.30	–	–	–	–
Assemble Model [5]	95.29	–	–	–	–
ANN (4-layer) [6]	92.39	–	–	–	–

1) Superiority Over Traditional Machine Learning Models: Classical machine learning models, such as Decision Tree (DT) and Random Forest (RF), achieved limited accuracy of around **74%**, struggling to model complex feature relationships. In contrast, the Voting Classifier delivered a **23%** improvement in accuracy (**97.02%**) and a ROC-AUC of **99.60%**, showcasing its enhanced discriminative power. Similarly, the Stacking Classifier demonstrated strong performance with an accuracy of **96.97%** and a ROC-AUC of **99.57%**.

2) Advantage Over Deep Learning Models: Deep learning models, including Artificial Neural Networks (ANN), Convolutional Neural Networks (CNN), Hybrid CNN+LSTM, and Multi-Layer Perceptrons (MLP), exhibited competitive but inferior performance. ANN achieved **91.78%** accuracy with a recall of **97.80%**, but lacked the precision and ROC-AUC of the Stacking Classifier. CNN, more suitable for image-based tasks, struggled with structured data, achieving only **74%** accuracy. Hybrid CNN+LSTM improved upon standalone methods with **95.90%** accuracy and a ROC-AUC of **98.90%**, while MLP models optimized with RMSProp (**94.98%** accuracy) and AdaMax (**94.30%** accuracy) demonstrated promise but fell short in recall and F1 score.

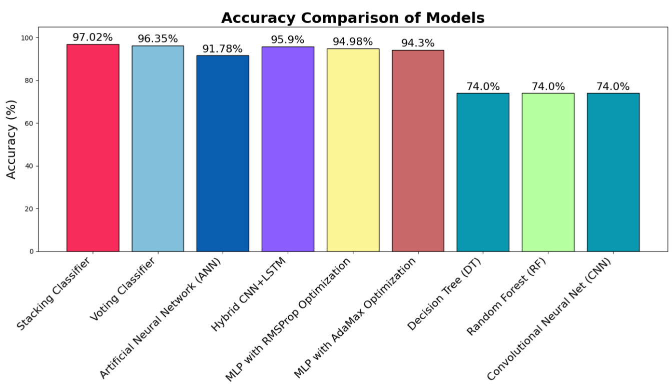

Fig. 7: Accuracy comparison of our proposed models with Baseline Models

The Stacking Classifier consistently achieved superior results, with **97.02%** accuracy, a ROC-AUC of **99.60%**, and higher recall **96.80%** and F1 score than hybrid deep learning methods. These findings underline the robustness and reliability of ensemble learning, establishing it as a superior approach to both traditional and deep learning techniques for stroke prediction.

V. Conclusion

This research identifies efficient hybrid ensemble learning models for stroke prediction, addressing challenges such as class imbalance and feature variability in medical data. The Voting Classifier achieved superior performance, with an accuracy of **97.02%** and a ROC-AUC of **99.60%**, while the Stacking Classifier achieved an accuracy of **96.35%** and a ROC-AUC of **99.09%**. Both models consistently outperformed traditional machine learning methods and deep learning approaches, including ANN and CNN+LSTM.

The proposed hybrid models are not only highly accurate but also computationally efficient, a key advantage over deep learning methods that often require substantial resources for training and deployment. This makes hybrid approaches smarter and more practical for real-world applications. By combining complementary predictions from base classifiers, these models offer robust and reliable performance, paving the way for effective clinical decision support. Future work could enhance these models by incorporating diverse data sources and patient populations, supporting early diagnosis and reducing stroke-related mortality.

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